PATRICK BARRY

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

- Bachelor of Science, Aerospace Engineering, May 2025 (Final GPA: 4.0)

Relevant Coursework: Spacecraft Flight Dynamics, Control System Analysis & Design, Dynamics & Controls Lab, Robotics and Autonomy, Numerical Analysis

- Master of Science, Aerospace Engineering, May 2026 through BS/MS Honors Program (Final GPA: 4.0)

Relevant Coursework: Optimal Guidance and Control, Optical Navigation, Kalman Filtering, Orbital Mechanics, Machine Learning, Dynamics of Mechanical Systems

SKILLS

Software: ROS, Linux, Git, GMAT, SolidWorks, Siemens NX, Microsoft Office
Programming Languages: MATLAB/Simulink, C, C++, Python, Java, CUDA
Interpersonal: Communication, Leadership, Teamwork, Organization

ACHIEVEMENTS

Eagle Scout, July 2020

PUBLICATIONS

Barry, Zheng, Caillet, and Reynolds. *Development of an LQR Controller for a Jet Vanes Thrust-Vectored Rocket.* AIAA ARC, 2025. https://arc.aiaa.org/doi/10.2514/6.2025-99074
Zheng, Barry, Ambati, and Vadia. *Integrated State Estimation Development for Actively Controlled Rockets.* AIAA ARC, 2025. https://arc.aiaa.org/doi/10.2514/6.2025-99521

EXPERIENCE

SpaceX
GNC Engineer, Starshield, June 2026 – Present

- Develop beam planning and tracking algorithms

Johns Hopkins University Applied Physics Laboratory, Laurel, MD
Guidance, Navigation, and Control Intern, May 2025 – August 2025

- Developed a robust controller analysis toolset with MATLAB/Simulink to compute stability margins and performance metrics across launch vehicle trajectories given model uncertainty
- Implemented a Kalman filter to perform post-flight error analysis of an inertially guided launch vehicle based on an inertial measurement unit (IMU) error model and measurements along the trajectory
- Designed and ran experiments to estimate error coefficients within a navigation-grade accelerometer, including high g-level vibration table and centrifuge testing in space-grade laboratory facilities
- Processed experimental data with digital filters and Fourier transforms to estimate accelerometer error coefficients with an Unscented Kalman Filter (UKF)

Dynamics and Control Systems Laboratory, Georgia Institute of Technology, Atlanta, GA
Undergraduate Research Assistant, Autonomous Racecar Group, January 2024 – December 2024

- Wrote a Model-Predictive Path Integral (MPPI) nonlinear control algorithm for the F1Tenth racecar platform
- Implemented controller software in Python using PyTorch library for GPU code acceleration; integrated code with the racecar's existing driving framework using Robot Operating System (ROS)
- Enhanced algorithm for a 13x controller loop speed increase from 3 Hz to 40 Hz
- Wrote simulation and accompanying visualization to fine-tune the controller cost function
- Performed extensive testing in an open laboratory environment to evaluate system performance

The Aerospace Corporation, Chantilly, VA
Vehicle Design and Analysis Intern, May 2024 – August 2024

- Modeled spacecraft concept designs in a concurrent engineering tool to satisfy mission architectures
- Collaborated with subject matter experts to assess feasibility of existing spacecraft designs
- Examined satellite Preliminary and Critical Design Reviews to characterize previous designs and assessed how modifications would affect the system's size, mass, and power
- Algorithmically analyzed attitude control and propulsion subsystems to balance performance and cost
- Presented study findings to clients and delivered comprehensive documentation packages

Aerospace Systems Design Laboratory, Georgia Institute of Technology, Atlanta, GA
Undergraduate Research Assistant, Sensor Development and Deployment, August 2023 – December 2023

- Collaborated in small team to develop low-cost wireless sensors to measure various properties (temperature, occupancy, air quality, etc.) in campus buildings
- Implemented electrical current sensor software on a Raspberry Pi microprocessor
- Designed sensor printed circuit board (PCB) to manage power supply, data flow, and wireless communication

Mitchell Frank Tennis Academy, Annandale, VA
Tennis Instructor, Summers 2021, 2022, and 2023

- Selected and organized drills and games for children aged 4-14 to foster skill development and self-confidence in a fun environment
- Promoted to manager of primary location for summer 2023; responsible for oversight of multiple coaches and more than 50 students
- Directed fitness program in summer 2023 for elite players seeking college recruitment, promoting discipline and mental toughness to aid players in achieving their goals

Campus Recreation Center, Georgia Institute of Technology, Atlanta, GA
Wilderness Outpost Attendant, January 2023 – May 2023

- Prepared outdoor gear per customer specifications for personal trips and expeditions
- Staffed the front desk, answered questions, and registered patrons for outdoor trips
- Identified innovations solution to unique problems to ensure the Outpost's needs were met

PROJECTS

Planetary Supply Drop, Personal Project
C++ and Python Code Development, May 2025 to present

- Developing a launch-to-landing simulation of a payload delivery to any specified point on Earth using C++ with Python for visualizations
- Utilizing a Kalman filter to perform orbit determination of a satellite based on ground observations
- Derived a Linear Quadratic Regulator (LQR) controller to perform orbit corrections needed due to atmospheric drag, solar radiation pressure, and non-spherical Earth gravity effects
- Implemented a convex trajectory optimization algorithm for a thruster-equipped lander to perform a soft landing

Ramblin' Rocket Club, Georgia Institute of Technology, Atlanta, GA
Avionics & Controls Lead, August 2023 – May 2025

- Led team of 25+ students to develop avionics suites, software, and controllers for thrust-vectored rockets actuated by jet vanes and motor gimbaling
- Numerically linearized a nonlinear dynamic rocket model to produce trim points and state-space representation of the vehicle across the flight profile
- Developed comprehensive object-oriented MATLAB simulation code repository centered around a closed-loop, 6-DOF simulation with a Linear Quadratic Regulator (LQR) controller to follow a reference trajectory
- Performed Monte Carlo simulations to ensure flight success under diverse environmental conditions and system modeling errors
- Wrote and implemented state estimation algorithms for position, velocity, and attitude in C on an STM32 microcontroller
- Developed electronics for a static fire test system, including remote commands, motor ignition, and servo timing

NASA Lucy Student Pipeline Accelerator & Competency Enabler (L'SPACE)
Deputy Project Manager, January 2023 – May 2023

- Participated in the L'SPACE Mission Concept Academy, receiving mission development skills training from top NASA scientists and engineers
- Served as Deputy Project Manager on an 11-person virtual team to assemble a Preliminary Design Review for a rover mission to Mars' northern polar region
- Led communications between science and engineering subteams while developing mission specifics, including timeline and operations, while considering past missions and budget constraints